ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 7-Layer OSI-Style Stack Reference (Based on Roitman arXiv:2606.24937)

What actually runs under the hood of an autonomous AI agent in 2026?
Like the 7-layer OSI model, modern agents are a vertical stack — from GPU silicon (L1) & uv/PyTorch/JAX (L2) to Data/RAG (L3), Arch/RLVR (L4), MCP (L5), Harness² (L6) & A2A swarms (L7).

① TRACE THE STACK: Follow Layers 1 → 7 from silicon to swarms
② BUILD EACH LAYER: Top 3 pills = flagship 2026 primitives & tools
③ HARDEN PROD: Audit the red FAILURE MODES & TRAPS on the right

Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling. Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-07 23:16:50 AEDT (UTC+11:00)

OSI AGENT LAYER	CORE PRIMITIVES & ARCHITECTURE	FRAMEWORKS, PACKAGES & PROTOCOLS	SCALING, VERIFICATION & SECURITY	FAILURE MODES & TRAPS
OSI Analogy · PDU · Scope · Ch.	Hardware Physics · Math · Data Structures · Control Topologies	CLI Package Managers · DL Frameworks · Serving · Agent SDKs	Parallelism · Sandboxes · Evals & Oracles · Guardrails	What Breaks Here (Anti-Patterns)
LAYER 1 **PHYSICAL & SILICON** GPUs/TPUs/NPUs, HBM3e vs. SRAM hierarchy, CUDA/Triton kernels, and NVLink/InfiniBand fabrics. • OSI L1 Physical · PDU: FLOPs, HBM Bytes & Warps • Guide: Ch. 3.6, 2.1, 11.10-14, 30.5 (GPUs & Kernels).	**GPU / TPU / NPU Tensor Cores** **HBM3e vs. On-Chip SRAM Hierarchy** **FlashAttention IO-Aware SRAM Tiling** Streaming Multiprocessors (SMs) · Warp Schedulers (32-Thread SIMT) Arithmetic Intensity (FLOP/Byte) · Roofline Compute vs. Memory Bound FP16 / BF16 / FP8 / FP4 Formats · Online Softmax Rescaling (Safe Max) Unified Memory Architecture (UMA) · Wafer-Scale & SRAM-Only LPU Dies Register File (256KB/SM) · L1 Cache / Shared Memory · L2 Cache (50MB–128MB) HBM3e Bandwidth (3.3–8 TB/s) · SRAM Bandwidth (~20–30 TB/s) · Transformer Engine FP8/FP4 Stochastic Rounding · Tensor Memory Accelerator (TMA) · Systolic Array Matrix Multiply Memory Coalescing · Bank Conflict Avoidance · TDP Power & Liquid Cooling	**NVIDIA H100 / H200 / Blackwell B200** **Google TPU v5p / v6e Trillium / v7** **CUDA · OpenAI Triton · Apple Silicon M4** NVIDIA GB200 NVL72 Rack Scale · AMD Instinct MI300X / MI325X AWS Trainium2 & Inferentia2 · Groq LPU & Cerebras WSE-3 ASIC NVLink 4/5 (900–1,800 GB/s/GPU) · InfiniBand NDR/XDR (400–800Gb/s) cuBLAS / cuDNN / CUTLASS / FlashInfer · AMD ROCm / HIP & Apple Metal DGX / HGX 8-GPU Node Topology · GPUDirect RDMA / Storage · Mellanox ConnectX-7 / 8 NICs 80GB / 141GB / 192GB VRAM Tiers · Mac Mini M4 64GB–512GB Cluster · RTX 4090 / 5090 Local Workstation On-Device NPU (45–80+ TOPS) · Co-Packaged Silicon Photonics · Vera Rubin NVL144 (HBM4 Roadmap) Liger-Kernel Fused Triton Ops · ECC Memory Error Correction · NCCL / Gloo Collective Backend	**Model FLOPs Utilization (MFU > 45%)** **Prefill (Compute) vs. Decode (BW)** **Confidential Compute GPU TEEs** Prefill Phase (Matrix-Matrix Gemm) · Decode Phase (Matrix-Vector Gemv) Multi-Instance GPU (MIG) Slices · Rail-Optimized Fat-Tree Topology NVIDIA DCGM Hardware Health Check · Silent Data Corruption (SDC) Scans Nsight Compute & Systems Profiling · Hardware Root-of-Trust Attestation MFU = Achieved / Peak FLOPs · Hardware FLOPs Utilization (HFU) · Bytes/Param: 2B (BF16) vs 0.5B (FP4) 4-Model RLHF VRAM Multiplier · Optimizer State Memory (12B/param) · Activation Memory Checkpointing NVLink All-Reduce Bandwidth · Cross-Node All-to-All Bottleneck · Thermal Throttling Mitigation H100/B200 Secure Enclave (CC-On) · Power Usage Effectiveness (PUE) · Straggler GPU Detection	**HBM Memory Bandwidth Wall (Decode)** **CUDA Out-of-Memory (OOM) Crashes** **Silent Tensor Core Bit-Flip Corrupt** ▸ Low Decode MFU (<5% Compute Util) ▸ PCIe Host-to-Device Bottleneck ▸ InfiniBand Link Flaps & Stalls ▸ Straggler GPU Halting 1,000 Nodes • 4-Model RLHF VRAM Explosion • Thermal Throttling Under RL Rollouts • Unaligned Tensor Shapes (No TC) • Consumer GPU P2P Disabled Locks
LAYER 2 **ENV, PACKAGES, PYTHON & TENSOR** Package managers (`Homebrew`, `uv`, `conda`, `pip`, `venv`), `Python`, `PyTorch`, `JAX`, `vLLM`, and sandboxes. • OSI L2 Data Link · PDU: Wheels, Tensors & Paged KV • Guide: Ch. 1.14–15, 11.1–9, 21.5, 22.1–9 (Env & Runtime).	**Virtual Environments, Locks & Sandboxes** **Autograd Computation Graphs & JIT** **PagedAttention KV & Continuous Batch** POSIX PTYs, Processes & Asyncio Loop · ABI / manylinux Wheels & uv.lock OS Namespaces, cgroups & Seccomp · Dynamic Autograd DAG & torch.compile PagedAttention Virtual Memory KV · RadixAttention Automatic Prefix Cache Speculative Decoding (Draft + Verify) · 3D Parallelism (DP/TP/PP/SP) & ZeRO pyproject.toml (PEP 517 / 621) · Content-Addressable Global Cache · Hardlink / Reflink Env Cloning LD_LIBRARY_PATH / DYLD RPATH · torch.compile (TorchInductor/Dynamo) · JAX jit / vmap / pmap / shard_map XLA HLO Graph Fusion & CUDA Graphs · Disaggregated Prefill-Decode Routing · Chunked Prefill & Guided Decoding KV-Cache FP8 / INT4 Quantization · Container UnionFS / OverlayFS Layers · Hardware Virtualization (KVM / HVF)	**Homebrew (brew) · uv · conda · pip · venv** **Python 3.12+ · PyTorch 2.x · JAX · MLX** **vLLM · SGLang · llama.cpp · Docker · E2B** Homebrew (brew) & System Toolchains · uv (uv venv / uv run / uvx / uv pip) conda / Miniforge / mamba / Pixi / Poetry · pip · venv · virtualenv · pipx PyTorch 2.x · JAX / Flax · Apple MLX · vLLM · SGLang · TensorRT-LLM · Ollama Docker / OCI & NVIDIA Container Toolkit · Rust (PyO3/Maturin) & TypeScript/Bun brew install uv git ripgrep fd tmux · uv lock & uv sync --frozen · environment.yml & conda-lock pip install --no-build-isolation · DeepSpeed ZeRO & PyTorch FSDP2 · llama.cpp (GGUF) & ExLlamaV2/v3 Outlines & XGrammar (JSON/CFG) · Ray Serve & NVIDIA Triton Server · ONNX Runtime & Safetensors Zero-Copy Nix / Devcontainers / Direnv · Slurm / Kubernetes (K8s) / Ray · Pydantic v2 (Rust Schema Core)	**E2B / Firecracker MicroVM Sandboxes** **OS Sandboxing (Bubblewrap / Seatbelt)** **TTFT / TPOT SLAs & Async RL Sync** gVisor User-Space Kernel Isolation · macOS sandbox-exec & Linux bwrap OpenEnv Standardized Gym Reset/Step · Air-Gapped VLAN & Egress Firewall Scoped Ephemeral OAuth / API Tokens · NCCL Weight Broadcast (Trainer->vLLM) Per-Tenant LoRA Adapter Hot-Swapping · Grammar-Constrained 100% Valid JSON env.reset() / env.step(action) API · SWE-bench Docker Repo Images · OSWorld / WebArena Live Desktop VMs TTFT < 200ms · TPOT < 15ms/tok · Goodput (SLO-Compliant Tok/s) · In-Flight Weight Update (Zero Stall) S-LoRA / Punica Multi-Adapter Batch · Pyodide / Wasm Code Execution · Secret Scrubbing in Env Vars SBOM & pip-audit Vulnerability Scan · Signed Container Attestation (Cosign) · Deterministic Seed Mode (cuDNN)	**Python Dependency & CUDA ABI Hell** **Unsandboxed Host RCE & rm -rf /** **KV-Cache Fragmentation & Preemption** ▸ nvcc vs. torch.version.cuda Mismatch ▸ Global System Python Pollution ▸ pip install Breaking flash-attn ABI ▸ Typosquatted PyPI / npm / Skill Packs • Agent Leaking .env & SSH Keys • 60–80% Contiguous KV Memory Waste • RL Rollout Generation Bottleneck • torch.compile Graph-Break Thrashing
LAYER 3 **DATA, TOKENIZATION & RAG** Pre-training/SFT/RLVR data curation, synthetic trajectories, BPE tokenizers, vector DBs, and Agentic RAG. • OSI L3 Network · PDU: Token IDs, Embeddings & Chunks • Guide: Ch. 1.2, 3.2–5, 4.3, 16, 18.4, 30.2 (Data & RAG).	**BPE Tokenization, Packing & Loss Masking** **SFT, Preference & RLVR Data Curation** **Embeddings, Vector HNSW & Hybrid RAG** Byte-Level BPE / SentencePiece / Tiktoken · Sequence Packing & Position-ID Reset Completion-Only Loss Masking (Labels=-100) · Chosen vs. Rejected Preference Pairs Synthetic Reasoning & Trajectory Gen · Bi-Encoder (HNSW) vs. Cross-Encoder Hybrid RAG (Dense + BM25 + RRF Fusion) · GraphRAG Entity Summaries & REFRAG MinHash LSH & Exact Substring Dedup · FastText / KenLM Quality Filtering · Decontamination (N-gram / Embed Match) ChatML Special Tokens (<|im_start|>) · Rejection Sampling (Best-of-N Filter) · Execution-Verified Code/Math Traces Contextual Chunk Headers (Anthropic) · Late Interaction ColBERT Token MaxSim · Search-R1 RL Query Reformulation Self-RAG Critique Tokens (ISREL/ISSUP) · Corrective RAG (CRAG) Web Fallback · Matryoshka Representation (MRL)	**HF Datasets · Datatrove · NeMo Curator** **Tiktoken · SentencePiece · HF Tokenizers** **Qdrant · Milvus · pgvector · Neo4j** FineWeb-Edu · Dolma · The Stack v2 · Apache Arrow · Parquet · WebDataset Docling · Unstructured · Marker PDF · Tree-sitter AST Codebase Chunking BGE-M3 · Cohere Embed v3 & Rerank v3 · FAISS · LanceDB · Chroma · Weaviate LlamaIndex · Haystack · LightRAG · Distilabel · Curator Synthetic Suites Polars · DuckDB · PyArrow ETL · Ray Data Distributed Streaming · MosaicML StreamingDataset (MDS) OpenMathReasoning & NuminaMath · SWE-Gym & SWE-smith Trajectories · ShareGPT & OpenAI Harmony Formats Firecrawl · Jina Reader Markdown · Graphiti & Cognee Knowledge Graphs · SQLite-vec & DuckDB-VSS Local DVC & LakeFS Data Version Control · Cleanlab Automated Label Audit · Argilla Human-in-the-Loop Curation	**RAG Triad Evals (Faithfulness / Recall)** **Benchmark Decontamination & PII Scrub** **Document-Level RBAC & Injection Scans** Ragas & TruLens Groundedness Oracles · NDCG@k · MRR · Recall@k Retrieval Audits Pre-Indexing ACL / RBAC Permission Tags · Regex + Presidio PII & Secret Scrubbing Canary String & N-Gram Contamination · Synthetic Data Diversity & Entropy Audit Citation Span-Locking & Provenance · Indirect Prompt Injection Sanitization LLM-as-a-Judge Data Quality Scoring · Perplexity Outlier Filtering (KenLM) · Topic & Difficulty Curriculum Mix Needle-in-a-Haystack (NIAH) & RULER · FRAMES & LongBench Multi-Hop Eval · Hallucination Span (LettuceDetect) Poisoned Document Backdoor Scans · Copyright & SPDX License Filtering · Cross-Tenant Vector Namespace Auth Embedding Drift & Stale Index Check · Tokenizer Fertility (Tok/Word) Audit · Zero-Retention Enterprise Pipelines	**Garbage-In Synthetic Model Collapse** **Naive Fixed-Size Mid-Table Chunking** **Indirect Prompt Injection via RAG** ▸ Training on Prompt Tokens (No -100 Mask) ▸ Test-Set Contamination Mirage ▸ Cross-Row Table & Function Splitting ▸ Unfiltered Vector RAG Leaking Docs • Chat Template EOS/BOS Token Mismatch • Tokenizer Glitch & Byte-Fallback Bloat • Unverified Synthetic Hallucinations • Hallucinated Citations on Unread Text
LAYER 4 **MODEL ARCHITECTURE & RLVR** Transformer/MoE/hybrid architectures, SFT, PEFT adapters (`LoRA`), preference tuning (`DPO`), and `RLVR` (`GRPO`). • OSI L4 Transport · PDU: Logits, Adapters & CoT • Guide: Ch. 1, 2.2, 3–10, 12–13, 30.1–3 (Models & RL).	**Transformer · Sparse MoE · GQA / MLA** **Hybrid Linear Attention (Mamba/DeltaNet)** **PEFT (LoRA/DoRA) & RLVR (PPO/DPO/GRPO)** Pre-Norm RMSNorm · SwiGLU · RoPE/YaRN · Grouped-Query (GQA) & Latent (MLA) Sparse MoE Top-K & Shared Experts · Gated DeltaNet / Mamba-2 SSM Hybrid LoRA · QLoRA (NF4) · DoRA · PiSSA · Preference Tuning: PPO · DPO · KTO Critic-Free RLVR: GRPO · Dr.GRPO · DAPO · Harness Distillation (Harness-Zero) Multi-Head Latent Attention (MLA) · Aux-Loss-Free MoE Load Balancing · rsLoRA (alpha/sqrt(r)) · QLoRA NF4 Zeroth-Order MeZO Forward-Only Tune · Bradley-Terry RM & Process RM (PRM) · IPO · ORPO · SimPO · Online DPO RLCR Calibrated Brier Score Reward · On-Policy Self-Distillation (OPSD/RLCSD) · Test-Time Compute (Best-of-N / MCTS) Cold-Start Pure-RL Emergence (R1-Zero) · MergeKit (SLERP / TIES / DARE) · AdamW vs. Muon Matrix Orthogonal	**GPT-5.4 / o3 · Claude 4.6/5 · Gemini 3** **DeepSeek-R1/V3.2 · Qwen 3 · Llama 4** **HF TRL · PEFT · verl · OpenRLHF · Unsloth** TRL (SFTTrainer / DPOTrainer / GRPO) · verl (HybridEngine RLHF/RLVR at Scale) OpenRLHF (Ray + vLLM Distributed RL) · Unsloth (2x Speed Triton Backprop) Axolotl & LLaMA-Factory YAML Suites · HuggingFace Transformers & PEFT Prime Intellect verifiers & OpenR1 · Torchtune & NeMo Aligner DeepSeek-R1-Distill (7B-70B Weights) · Qwen3-235B-A22B & QwQ-32B · Claude Sonnet 4.5 / Opus 4.6 / 5.5 OpenAI GPT-5.4-Codex & o4-mini · Gemini 2.5 / 3.1 Pro & 3.8 Flash · Mistral Large 3 & Codestral bitsandbytes (NF4) · AWQ · GPTQ · Lean 4 / SymPy Formal Math Verifiers · MergeKit CLI Checkpoint Merger WandB / TensorBoard / MLflow Logs · vLLM Rollout Worker Integration · BitNet.cpp 1.58-Bit Ternary Weights	**Verifiable Execution Oracles (RLVR)** **KL-Divergence & Entropy Regularization** **Constitutional AI & Safety Classifiers** Deterministic Unit-Test / Math Graders · DAPO Clip-Higher & Dynamic Sampling Multi-Objective Reward Combination · Listwise Rank-Based Reward Models Instruction Hierarchy Privilege Training · Deliberative Alignment in CoT Traces Activation Steering & SAE Circuit Probes · GCG / AutoDAN / Best-of-N Red-Teaming KL Penalty beta * D_KL(pi_theta || pi_ref) · PPO Clipping epsilon in [0.1, 0.28] · GRPO Group Size G = 8..64 Rollouts Pass@k vs. Majority@k Scaling · Length-Controlled Win-Rate Audits · Catastrophic Forgetting Evals Sleeper Agent Backdoor Detection · Sandbagging & Alignment Faking Audits · Machine Unlearning (RMU / Grad Ascent) CBRN / Cyber ASL-3/4 Threshold Gates · Contamination-Free LiveBench / HLE · Calibration Expected Error (ECE)	**Reward Hacking & Length Inflation** **Entropy Collapse & Mode Collapse** **Catastrophic Forgetting / Alignment Tax** ▸ GRPO Zero-Variance Advantage (All 0/1) ▸ DPO Likelihood Displacement (Both Drop) ▸ MoE Routing Collapse (Expert Starve) ▸ Naive INT4 Quant Breaking Reasoning • Sycophancy & Delusional Validation • Emergent Misalignment from Narrow SFT • Over-Refusal on Benign Triggers • FP16 Loss-Scaling Underflow / NaNs
LAYER 5 **CONTEXT, MEMORY, TOOLS & MCP** Context budgeting, 4-tier agentic memory (`Sleep-Time Compute`), function calling, `MCP`, and `SKILL.md`. • OSI L5 Session · PDU: Context Windows & JSON-RPC • Guide: Ch. 17, 18.2–3, 23, 24 (Memory, Tools, MCP & Skills).	**Context Engineering & Token Budgeting** **4-Tier Memory + Sleep-Time Dreaming** **Tool Calling, MCP Bus & Progressive Skills** Working Memory (Active Context Window) · Episodic Memory (Past Experience Logs) Semantic Memory (Entity/Fact Triples) · Procedural Memory (SKILL.md / Weights) Sleep-Time Offline Consolidation (Dream) · JSON-RPC 2.0 Host-Client-Server Bus Progressive Disclosure Skill Loading · Observation Masking & /compact Hygiene System / History / RAG / Tool Scratchpad · Write / Read / Reflect / Consolidate Ops · Ebbinghaus Temporal Memory Decay Proactive Context Pre-Fetching · KV-Cache Prefix Pinning (90% Discount) · Parallel Tool-Call Batch Execution Strict JSON Schema Tool Definitions · Code-Act (Python Execution as Tool) · Dynamic Tool Search (Top-K Schemas) Rolling Summary & Turn Truncation · Subagent Forked Context Isolation · Scratchpad & Todo.md State Offload	**Model Context Protocol (MCP / .mcp.json)** **Agent Skills (SKILL.md Folder Standard)** **Letta (MemGPT) · Mem0 · Zep · LangMem** MCP Primitives: Tools, Resources, Prompts · MCP Transports: stdio & Streamable HTTP FastMCP Python / TypeScript SDKs · SKILL.md YAML Frontmatter + Instructions skills/scripts/ & skills/references/ Dirs · Letta (MemGPT) OS-Paged Memory Blocks Mem0 · Zep · Graphiti Temporal Memory · Linux Foundation AAIF MCP Registry MCP Sampling & Roots Client Callbacks · MCP Elicitation User Confirmation · .claude/skills/ & .agents/skills/ Paths agentskills.io Open Specification · LangMem Semantic/Episodic Store · Cognee & Memary Memory Engines Redis / Postgres Agent Session Stores · OpenAPI-to-MCP Auto-Generators · Browserbase / Playwright MCP Servers GitHub / Slack / Postgres MCP Servers · Anthropic & Cursor MCP Marketplaces · Smithery & Glama MCP Registries	**MCP OAuth 2.1 Scopes & Tool Signing** **/compact Hygiene (<65% Context Cap)** **Memory Contradiction & Decay Audits** Human-in-the-Loop Destructive Tool Gate · Tool Schema Token-Bloat Minimization Memory Contradiction & Stale-Fact Purge · LOCOMO & LongMemEval Memory Evals MCP Server Code-Signing Verification · Least-Privilege Read-Only Connectors OX Security stdio RCE Hardening · Cross-Tenant Memory Isolation Context Utilization < 65% Rot Threshold · Lost-in-the-Middle U-Curve Placement · BFCL (Berkeley Function-Calling) Eval ToolBench & API-Bank Multi-Tool Eval · Episodic Replay Buffer Deduplication · Tool Output Truncation & Pagination Skill Static Analysis Before Install · Egress Domain Whitelisting per Tool · OAuth 2.1 PKCE Token Refresh Flow Idempotency Keys for Mutating Tools · Audit Logging of All JSON-RPC Calls · Canary Tokens in Memory Stores	**Context Rot & Lost-in-the-Middle** **MCP Tool Bloat (100+ Schemas)** **Malicious Skills & MCP Rug-Pulls** ▸ 100+ MCP Tool Schemas Blowing Context ▸ ClawHavoc Malicious SKILL.md Payloads ▸ MCP Rug-Pull & Tool Shadowing Attacks ▸ Memory Poisoning Across Sessions • Stale Episodic Memory Overriding Code • stdio MCP Server Spawning Host Shells • Malformed Tool-Call JSON Loops • Unbounded Tool Output Flooding Window
LAYER 6 **AGENT HARNESS, LOOP & GRAPH** The CPU/OS of an agent: `CLAUDE.md`/`AGENTS.md`, Loop Engineering, Graph Engineering (`LangGraph`), and `Harness²`. • OSI L6 Presentation · PDU: State Transitions & Diffs • Guide: Ch. 18, 19, 20, 21, 27 (Harness, Loops & Graphs).	**Agent Harness (Prompt+Tools+State+Loop)** **Loop Engineering (Observe→Decide→Act)** **Graph Engineering (Typed StateGraphs)** ReAct · Reflexion · Plan-and-Execute · 5 Loop Primitives (State/Step/Check/Stop) Typed State Schema & Reducer Functions · Conditional Edges & Cyclic Graph Routing Fan-Out / Fan-In Parallel Send() API · Recursive Agent Harnessing (Harness²) Spec-Driven Development (Spec→Plan→Task) · Cognitive Escalation & Uncertainty Handoff pi(a_t | s_t) Inference-Time Policy Loop · LATS (Language Agent Tree Search) · LLM Compiler Parallel DAG Execution BDI (Belief-Desire-Intention) Hybrid · Checkpointer Time-Travel & Replay · Human-in-the-Loop Interrupt() Gates Stagnation & Repeat-Action Detection · Subagent Context Isolation (Forked Call) · Red-Green-Refactor TDD Agent Cycle Exponential Backoff & Schema Self-Repair · Token/Cost/Iteration Termination Caps · Dynamic Model Routing (Fast vs. Reasoner)	**Claude Code CLI · Codex CLI · OpenCode** **Cursor · Windsurf · Cline / Roo · Aider** **LangGraph · OpenAI Agents SDK · DSPy** CLAUDE.md & AGENTS.md Repo Constitutions · .cursor/rules & .github/copilot-instructions GitHub Spec Kit & obra/superpowers TDD · LlamaIndex Workflows (Event-Driven) Smolagents (CodeAct Python-as-Action) · PydanticAI Type-Safe Agent Harness OpenHands (OpenDevin) & Block goose · Semantic Kernel & Haystack Agents LangGraph PostgresCheckpointer · DSPy MIPROv2 Prompt/Weight Compiler · Claude Agent SDK & Computer Use API Browser-Use & Playwright DOM Harnesses · Vercel AI SDK Core (generateText/maxSteps) · Guardrails AI & NeMo Guardrails Instructor & BAML Structured Outputs · Litellm Unified Model Router · Git Pre-Commit & CI Harness Hooks Repomix / Gitingest Codebase Packers · LangSmith · Arize Phoenix · Langfuse · OpenTelemetry (OTel) GenAI Spans	**Decoupled Verification Oracles (No Self-Grade)** **SWE-bench Verified/Pro & Terminal-Bench** **pass^k Reliability & Trajectory Tracing** Property-Based Testing (Hypothesis/Kani) · Mutation Testing (Stryker / mutmut) Gates Immutable CI Tests Locked From Agent Write · Trajectory Tool-Call Exact/Prefix Match LLM-as-a-Judge Pairwise Rubric Grading · Cost-per-Resolved-Task Efficiency Metric PyRIT & Garak Adversarial Red-Teaming · LlamaFirewall Prompt/Tool Guardrails pass^k = (pass@1)^k Consistency Metric · METR 50%-Reliability Horizon (2h-30h) · GAIA · OSWorld · WebArena · TAU-bench Scale SWE-bench Pro (Private Repos) · Step Efficiency (Optimal / Actual Steps) · Deterministic Mock-Tool Replay Harness Shadow-Mode Traffic Counterfactual Eval · Circuit-Breaker Spend & Loop-Depth Caps · Canary Token Exfiltration Tripwires Git Diff AST Security Linter (Semgrep) · Automated Rollback on Regression · EU AI Act / SOC2 Immutable Audit Logs	**Agent Deleting / Mocking Failing Tests** **Unbounded Loop Burn & Action Stagnation** **God-Node Anti-Pattern & State Bloat** ▸ Coupled Agent Rewriting Its Own Grader ▸ 0.95^20 = 35.8% Compounding Step Decay ▸ Poisoned .cursor/rules or GitHub Issues ▸ CVE-2025-53773 IDE Config Auto-Approve RCE • Swallowing Exceptions in Tool Wrappers • Appending Raw 50KB HTML to Graph State • Premature Termination Before Verification • Sycophantic Plan Pivot on User Pushback
LAYER 7 **MULTI-AGENT MESH, A2A & UI** Multi-agent topologies (`OWC`), inter-agent protocols (`A2A`/`ACP`), parallel worktree swarms, and Agentic/Generative UI. • OSI L7 Application · PDU: A2A Tasks & GenUI Streams • Guide: Ch. 21.6, 25, 26, 28, 30.10/18 (A2A, MAS & Agentic UI).	**Orchestrator-Worker-Critic (OWC) Pattern** **Multi-Agent Topologies (Hierarchical/Mesh)** **Generative UI & Shared State Sync** Centralized Supervisor vs. Decentralized · Hierarchical Tree (Squads of Squads) Debate / Society-of-Mind Consensus · Blackboard Shared-Memory Coordination Contract-Net Task Bidding & Auctions · Parallel Git-Worktree Swarm Isolation A2A (Horizontal) × MCP (Vertical) Dual Bus · Human-on-the-Loop Escalation & Approval Agent Card (/.well-known/agent.json) · Task Lifecycle (Submitted→Working→Done) · Multi-Agent MDP & CTDE MARL Training Credit Assignment (Shapley / Counterfactual) · Role Specialization & Persona Diversity · Byzantine Fault-Tolerant Critic Voting Pub/Sub Asynchronous Event Choreography · Process-Centric Enterprise State Workflows · Server-Driven UI Component Streaming Optimistic UI Updates & Diff Previews · Ambient OS Cross-App Delegation · Organizational Digital-Twin Simulation	**Google A2A Protocol · IBM ACP · ANP** **CrewAI · AutoGen / AG2 · LangGraph Swarm** **CopilotKit · AG-UI · Vercel AI SDK UI** Google Agent-to-Agent (A2A) JSON-RPC/SSE · IBM BeeAI & Agent Communication Protocol Microsoft AutoGen v0.4 / Semantic Kernel · CrewAI Flows & Role-Based Crews Claude Code Subagents & Worktree Swarms · OpenClaw 24/7 Multi-Channel Agent Daemon Agent Gateways (Kong / Solo / LiteLLM) · Claude Artifacts · ChatGPT/Gemini Canvas AG-UI (Agent-User Interaction Protocol) · Vercel AI SDK useChat / streamUI Hooks · Chainlit · Streamlit · Gradio Agent UIs OpenAI ChatKit & Realtime WebRTC Voice · LangGraph Platform Cloud & Studio UI · Camel-AI Workforce & MetaGPT SOPs OpenAI Swarm / Agents SDK Handoffs · A2A Artifact & Push-Notification Webhooks · React Server Components (RSC) GenUI Slack / Discord / WhatsApp Bot Bridges · Linear / GitHub PR Async Agent Teammates · 1-Click MCP & A2A Enterprise Registries	**Agent Gateway (mTLS, Auth, FinOps Caps)** **Cross-Agent Trust & Cryptographic KYA** **Independent Critic & Red-vs-Blue Swarms** Zero-Trust Agent Identity (SPIFFE / KYA) · OAuth 2.1 On-Behalf-Of (OBO) Token Flow Per-Swarm Token & Dollar Circuit Breakers · Anti-Collusion & Decorrelated Critic Models Cross-Repo API Contract Locks (OpenAPI) · Deadlock, Livelock & Cycle Detection Provenance Watermarking (C2PA / SynthID) · Strict Liability & Action Audit Ledgers Communication Overhead O(N^2) Pruning · Multi-Agent Judge Independence Check · Cross-Org Data-Loss Prevention (DLP) Swarm Amplification Factor Monitoring · Canary Deployment of Agent Topologies · Chaos Engineering (Injecting Rogue Agents) XBOW / Cybench / T3MP3ST Cyber Ranges · Lean 4 / TLA+ Formal Protocol Verification · Human Escalation SLA Latency Tracking Anti-Sycophancy & Anti-Psychosis Filters · GDPR / HIPAA Cross-Agent Consent Scopes · Kill-Switch Broadcast Across Swarm Nodes	**Cascading Hallucination & Swarm Echo** **O(N²) Chatty Agent Token Explosion** **Cross-Agent Confused Deputy & Escalation** ▸ 7-Persona Roleplay Chatter Burning 15x Cost ▸ Homogeneous Critic Blindly Agreeing ▸ Supervisor Bottleneck & Context Overflow ▸ Merge Conflicts Across Parallel Worktrees • Rogue Subagent Bypassing Parent Scopes • Infinite Ping-Pong Handoff Deadlocks • Human Escalation SLA Latency Tracking • UI Stream Desync & Phantom Tool States